

1. Executive Summary

The **Azure File Sharing Web App** is a modern, secure, and scalable full-stack web application that allows users to upload, manage, and securely share files of any type using Microsoft Azure infrastructure.

Built with **React + Express.js** and hosted on **Azure App Service**, the application uses **Firebase Authentication** for Google sign-in and **Azure Blob Storage** with **SAS tokens** for secure, direct file operations. The system ensures complete user data isolation and provides time-bound secure sharing links.

Key Achievements:

- Successful deployment on Azure
 - Google login working
 - File upload, download, delete working
 - Share link generation implemented
 - Clean, responsive dark UI
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2. Project Objectives

- Provide a secure platform for users to store and share files using cloud services.
 - Implement **user isolation** so each user can only access their own files.
 - Use **direct browser-to-blob** uploads via SAS tokens for better performance and security.
 - Enable **time-limited secure sharing** with optional password protection.
 - Make the application production-ready using Azure best practices.
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3. Technology Stack

Frontend:

- React 19 + Vite
- TypeScript
- Tailwind CSS v4 + shadcn/ui

- React Dropzone, Lucide Icons, Sonner (toasts)

Backend:

- Node.js 20 + Express.js
- Firebase Admin SDK (token verification)

Cloud Services:

- **Azure App Service** (Hosting)
- **Azure Blob Storage** (File storage + SAS tokens)
- **Firebase Authentication** (Google Sign-In)

Deployment:

- GitHub + GitHub Actions (CI/CD)
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4. System Architecture

The application follows a **modern decoupled architecture**:

1. **Client Layer** → React SPA (Vite build)
2. **Authentication Layer** → Firebase Auth
3. **API Layer** → Express.js server running on Azure App Service
4. **Storage Layer** → Azure Blob Storage (private container)
5. **Security Layer** → Firebase ID Token + Azure SAS Tokens

Important Design Decisions:

- Files are stored with prefix: user@email.com/filename.ext
 - Uploads happen **directly** from browser to Azure Blob using temporary SAS tokens (no file passes through backend server)
 - All API calls are protected with Firebase token verification
 - Share links are temporary SAS read URLs with configurable expiry
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5. Key Features Implemented

Feature	Status	Description
Google Authentication	✓ Done	Firebase Auth + Easy login
File Upload	✓ Done	Drag & drop with progress
File Listing	✓ Done	Shows only user's files
Download	✓ Done	Generates temporary SAS URL
Delete	✓ Done	Secure deletion
Share Link Generation	✓ Done	Time-bound shareable links
Responsive UI	✓ Done	Dark theme, mobile friendly

6. Security Features

- Private Blob Container (no public access)
 - Short-lived SAS tokens (15 minutes for upload, 24 hours for sharing)
 - Firebase token validation on every API request
 - User-based file isolation
 - No sensitive credentials stored in frontend
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7. Deployment & CI/CD

- Code hosted on GitHub
 - Automatic build and deployment using **GitHub Actions**
 - Deployed on **Azure App Service** (Central India region)
 - Uses npm run build + node server/server.js
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8. Challenges Faced & Solutions

Challenge	Solution Implemented
Route ordering in Express	API routes placed before static + catch-all
BlobNotFound in share links	Added decodeURIComponent
Express 5 wildcard route error	Used middleware catch-all instead of *
Missing API routes	Added POST /api/share/:blobName
Container startup issues	Proper 0.0.0.0 binding + PORT env

9. Conclusion & Future Scope

The project has successfully achieved its core objective — building a secure, functional file sharing system on Azure using modern tools.

Future Enhancements (Phase 2):

- Email sharing (SendGrid / Azure Communication Services)
 - "Shared with me" section
 - File preview (images, PDF)
 - Password-protected shares
 - Analytics & usage tracking
 - Custom domain + HTTPS
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Project Status: Successfully Deployed & Functional **Live URL:** <https://yourfileshareapp2026-d7cpf0cehxambcdr.centralindia-01.azurewebsites.net>

End of Report